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A paleomagnetic and geochronologic study of the Majhgawan kimberlite, India: Implications for the age of the Upper Vindhyan Supergroup

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Abstract

A paleomagnetic and geochronologic investigation was undertaken on the Majhgawan kimberlite near Panna, Central India. Mineral $^{40}Ar/^{39}Ar$ ages on phlogopite separates from the kimberlite yield a mean age of 1073.5 ± 13.7 Ma (2σ). Paleomagnetic samples from the brecciated kimberlite yielded a mean direction with a declination of 45.3° and an inclination of −25.1° ($k = 37$, $\alpha_{95} = 9.3°$). When combined with directional data from an earlier study they yield a virtual geomagnetic pole at 36.8°N, 212.5°E (dp = 9.0°, dm = 16.6°). The new age for the Majhgawan kimberlite constrains the ages of Upper Vindhyan sedimentation (Bhander–Rewa) to less than ~1075 Ma.
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Keywords: Vindhyan; Paleomagnetism; $^{40}Ar/^{39}Ar$; Geochronology; Kimberlite

1. Introduction

The age of sedimentation within the Vindhyanchal basin is a controversy that has been disputed for over one hundred years (Venkatachala et al., 1996; Auden, 1933). While the age of the Lower Vindhyan Group is now well constrained, the Upper Vindhyan section has yet to be dated with any certainty. The onset of sedimentation in the Vindhyanchal basin commenced sometime after 1850 Ma based on U–Pb ages from underlying volcanic rocks (Deb et al., 2002). Geochronologic data from the Lower Vindhyan sequence (Fig. 1a; Table 1) are derived from U–Pb ages on porcellanites that yield 1628 ± 8 Ma, 1631.2 ± 5.4 Ma and 1630.7 ± 0.8 Ma (Rasmussen et al., 2002; Ray et al., 2002). Ages from the Rampur ash beds, below the Rhotas limestone, yield SHRIMP Pb–Pb ages of 1592 ± 12 and 1602 ± 10 Ma. Less precise Pb–Pb dates from carbonate yielded ages of 1599 ± 48 Ma (Sarangi et al., 2004) and 1601 ± 130 Ma (Ray et al., 2003) for the Rhotas limestone.

In comparison, the Upper Vindhyan sedimentary rocks lack reliable age constraints. Ray et al. (2003) reported a poorly defined Pb–Pb age on the Bhander Limestone of 750 Ma that they considered unreliable. Ray et al. (2003) also used Sr-isotopic data on the limestones to estimate a ~750 or 650 Ma age for the Bhander limestones; however, their Sr-isotopic data are also consistent with older and younger sections of the global isotopic curve. De (2003) reported 'Ediacaran-like'

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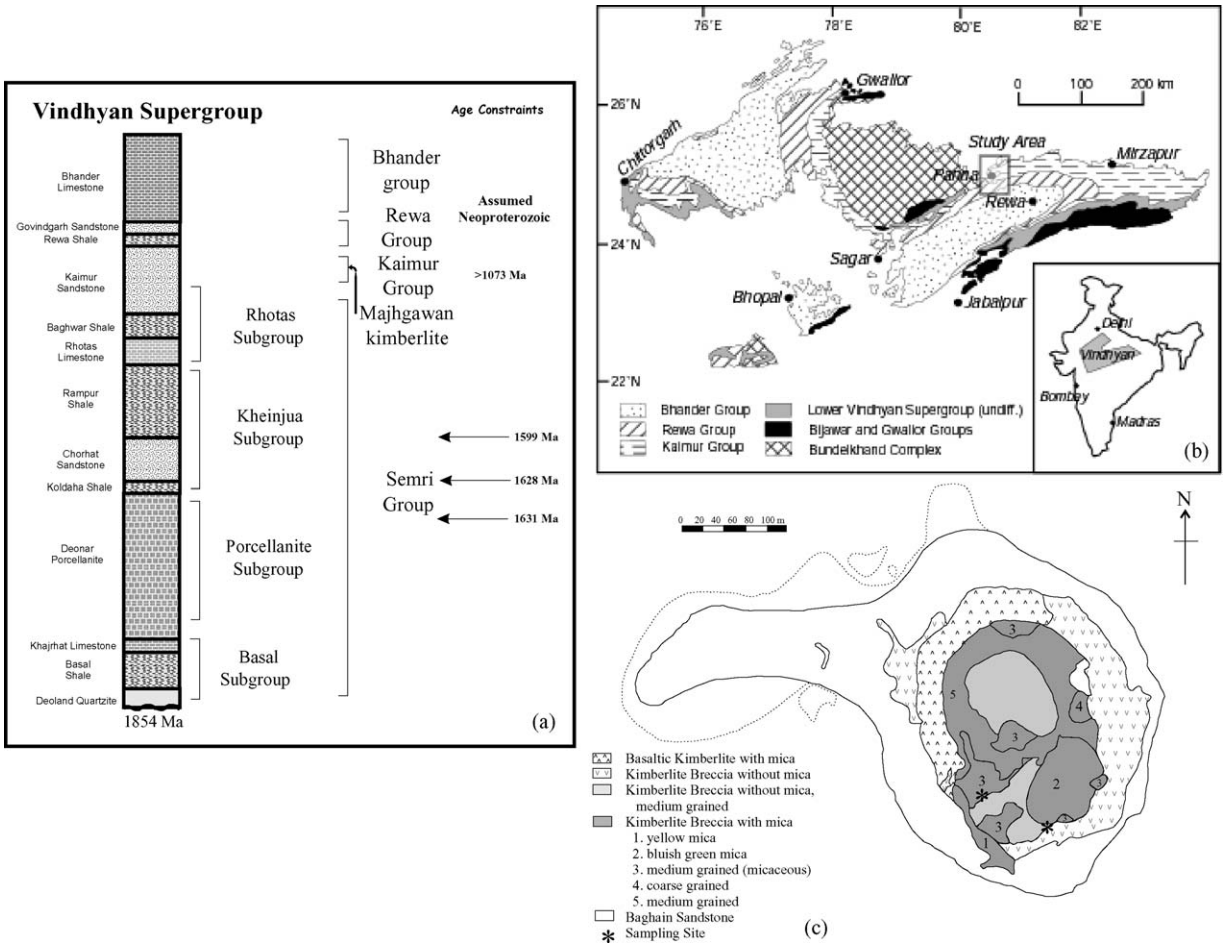


Fig. 1. (a) Generalized stratigraphic column of the Vindhyan Supergroup shown with radiometric age constraints. The Majhgawan kimberlite intrudes the Kaimur sandstone of the Upper Vindhyan. (b) Geologic map of the region including the Vindhyanchal basin and the study area near Panna, India. (c) Geologic map of the Majhgawan kimberlite (after Chatterjee and Rao, 1995). Sampling locations for paleomagnetism were taken from the basaltic kimberlite and the brecciated kimberlite. Samples from the Miller and Hargraves (1994) study were solely from the brecciated kimberlite.

organisms with very poor preservational characteristics from the Bhander limestone. The age range from these is consistent with the estimate by McElhinny et al. (1978) of an Ediacaran–Cambrian age for the Bhander–Rewa Groups. McElhinny et al. (1978) compared directions in the Bhander–Rewa rocks to known Cambrian rocks from the Salt Range in Pakistan. A more recent paper by De (in press) yields Ediacaran fossils with much better preservation in the Lakheri Limestone (Bhander Group). If the fossil find is confirmed by future work, then the Bhander Group is of Ediacaran age (~635–542 Ma; Condon et al., 2005; Jiang et al., 2003; Zhang et al., 2005).

The Majhgawan kimberlite, that intrudes Upper Vindhyan rocks (Fig. 1b and c), is important for obtaining a more reliable age for the basin due to its stratigraphic location (intruding the Kaimur sandstones). The reported ages of Majhgawan from previous studies range from 1630 ± 353 to 947 ± 30 Ma (see Table 1; also Auden, 1933). The kimberlite was dated by Crawford and Compston (1970) with mica to yield an age of 1116 ± 12 Ma. Two K–Ar whole rock and one mica determination yielded ages of 947 ± 30, 1170 ± 46 (whole rock, Paul et al., 1975) and 1056 Ma (mica, Crawford and Compston, 1970). Two recent ages of 1044 ± 22 and 1067 ± 31 Ma were reported using Rb–Sr systematics (Smith, 1992; Kumar et al., 1993 respectively).

The Majhgawan kimberlite is located in Madhya Pradesh, India (Fig. 1). It is one of several Proterozoic-age kimberlites/lamproites intruding the peninsular Indian crust. Other major bodies thought to be broadly consanguineous with the Majhgawan kimberlites are the

Table 1
Age constraints on Vindhyan sedimentary sequence

Stratigraphic layer	Method	Age (Ma)	Reference
Upper Vindhyan			
Bhavpura Shale (Bhander)	K–Ar	550	Crawford and Compston (1970)
Bhander Limestone	Pb–Pb	650	Ray et al. (2003)
Bhander Limestone	Sr-isotopes	750	Ray et al. (2003)
Bhander Limestone	Fossils	Ediacaran–Cambrian	De (2003, in press)
Kaimur Sandstone	K–Ar	940	Vinogradov et al. (1964)
Kaimur Sandstone	K–Ar	1071.6 ± 169.3	Srivastava and Rajagopalan (1988)
	F–T	1070.5 ± 160.4	Srivastava and Rajagopalan (1988)
Lower Vindhyan			
Rotasgarh Limestone	K–Ar	1400 ± 70	Vinogradov et al. (1964)
Rotasgarh Limestone	Pb–Pb	1601 ± 130	Ray et al. (2003)
Rohtas formation	Pb–Pb	1599 ± 48	Sarangi et al. (2004)
Rampur formation	K–Ar	1110 ± 60	Vinogradov et al. (1964)
Rampur formation	K–Ar	1124.5 ± 157.8	Srivastava and Rajagopalan (1988)
Rampur ash beds	Pb–Pb	1592 ± 12	
		1602 ± 10	
Porcellanites	U–Pb	1628 ± 8	Rasmussen et al. (2002)
		1631.2 ± 5.4	Rasmussen et al. (2002)
		1630.7 ± 0.8	Ray et al. (2002)
Kimberlites			
Majhgawan Kimberlite	Ar–Ar	1073 ± 13.7	This study
Majhgawan Kimberlite	K–Ar	1116 ± 12	Crawford and Compston (1970)
Majhgawan Kimberlite	K–Ar	947 ± 30	Paul et al. (1975)
Majhgawan Kimberlite	K–Ar	1170 ± 46	Paul et al. (1975)
Majhgawan Kimberlite	K–Ar	1056	Crawford and Compston (1970)
Majhgawan Kimberlite	Rb–Sr	1044 ± 22	Smith (1992)
Majhgawan Kimberlite	Rb–Sr	1067 ± 31	Kumar et al. (1993)
Wajrajur kimberlite	Rb–Sr	1350 ± 294	Paul (1979)
Wajrajur kimberlite	Rb–Sr	1116	Crawford and Compston (1973)
Lattavaram kimberlite	Rb–Sr	1505	Paul (1979)
Lattavaram kimberlite	K–Ar	933	Paul et al. (1975)
Muligiripalle kimberlite	U–Pb	1079	Miller and Hargraves (1994)

Lattavaram, Wajrakur, Narayanpet and Mulligiripalle kimberlites of the Dharwar craton (Miller and Hargraves, 1994; Haggerty and Birkett, 2004) and the Hinota pipe (Aravalli craton). Not all of these kimberlites have reliable age constraints. The Wajrakur kimberlite has ages ranging from 840 to 1350 Ma (Paul, 1979; Crawford and Compston, 1973; Paul et al., 1975). The Lattavaram kimberlite is dated between 933 and 1505 Ma (Paul, 1979; Paul et al., 1975). A U–Pb (perovskite) age of 1079 Ma was cited in Miller and Hargraves (1994) for the Muligiripalle kimberlite. The Lattavaram, Wajrakur and Mulligiripalle pipes have also been studied paleomagnetically by Miller and Hargraves (1994). The Wajrakur samples were unstable and yielded no useful data whereas the Lattavaram and Mulligiripalle pipes yielded similar paleomagnetic data, although the site statistics showed poor grouping and large errors. Miller and Hargraves (1994) also conducted a paleomagnetic study of Majhgawan and reported a paleomagnetic pole at 38.9°N, 216.5°E.

2. Geology of the kimberlite

The Majhgawan kimberlite intrudes Baghain sandstone in the upper Kaimur Group, which lies below the Bhander–Rewa sequence in the Upper Vindhyan basin (Fig. 1). The body is pear shaped with steeply dipping walls (80°) and its surface originally outcropped over an area of 500 m × 320 m (Chatterjee and Rao, 1995). Deformation of the Baghain sandstone is severe in the vicinity of the pipe. The Kaimur rocks have been shattered and tilted inward toward the pipe, but elsewhere in the region the Baghain sandstone and overlying rocks of the Bhander–Rewa Groups have very low dips (<10°) or are undeformed. The pipe has a variable lithology and is composed of concentrically arranged units. The

classification of the pipe is debated as it has characteristics of both a kimberlitic intrusion and an olivine lamporitic tuff. Chatterjee and Rao (1995) classify it as an intermediate between the two. The kimberlite is observed to have three intervals of intrusion: (1) deep-green brecciated kimberlite; (2) botryoidal zone of kimberlite contaminated by shale xenoliths; and (3) basaltic kimberlite. The pipe is mined for diamonds and also contains megacrysts of olivine, phlogopite, ilmenite, pyrope and enstatite with phologopite mica as a constituent of a serpentine and calcite matrix (Mukherjee et al., 1997). Phlogopite megacrysts can be placed into two distinct groups. Megacrysts are either rounded and commonly weathered or hexagonal and fresh (Mukherjee et al., 1997).

Intraformational conglomerates within the Rewa Group near Panna are diamond-bearing suggesting that the deposition of the Rewa Group post-dates intrusion of the kimberlite (Mathur, 1962, 1981). Rau and Soni (2003) examined the provenance of the diamondiferous conglomerates in the Rewa Group and concluded that the source for the diamonds in the Rewa Group may be something other than the nearby Majhgawan or Hinota pipes. No significant erosional surfaces are present in between the Kaimur, Rewa and Bhander Groups (Prasad, 1984).

3. Paleomagnetism

3.1. Paleomagnetic experiments

Samples were drilled in the field using a gasoline powered drill and oriented using magnetic and sun compasses. Sun compass readings were used to correct for the local declination and any rock magnetic interference. Limited outcrop and security concerns allowed us to collect only 10 samples from two sites in the brecciated kimberlites and basaltic kimberlites. Samples were then cut into standard specimens and stored in a magnetically shielded room at the University of Florida paleomagnetic laboratory. Preliminary samples were stepwise treated using thermal or alternating field demagnetization and after evaluation, a series of demagnetization steps was chosen. Alternating field demagnetization was conducted using a home-built AF-demagnetizer and with fields up to 100 mT. Thermal demagnetization was conducted up to temperatures of 600 °C with an ASC-Scientific thermal demagnetizer and all samples were measured in a ScT Cryogenic magnetometer. Principle component analysis (Kirschvink, 1980) was used to determine the best-fit lines for each sample.

3.2. Rock magnetic experiments

The susceptibility of each sample was measured before treatment on an Agico SI-3B bridge. In order to further characterize magnetic carriers, Curie temperature runs were conducted on several powdered samples using a KLY-3S susceptibility bridge adapted with a CS-3 heating unit. For this experiment, susceptibility is measured incrementally during heating and cooling of the samples. Isothermal remanence acquisition studies (IRM) were also conducted on select samples.

4. Geochronology

Samples of micaceous brecciated kimberlite were crushed, sieved and phlogopite separates were hand picked. The individual phlogopite samples were then ultrasonically cleaned and rinsed several times. The grains were examined again under microscope and any grains with weathered margins or impurities were discarded. The grains were then wrapped in aluminum foil and sent to Oregon State University's irradiation facility. Gas from the samples was stepwise treated using CO_2IR laser and measured on a MAP215-50 mass spectrometer. The flux monitor was GA1550 biotite (age 98.8 ± .5 Ma; Renne et al., 1998).

5. Results

5.1. Geochronology

The phlogopite micas contain roughly 10% K_2O (Mukherjee et al., 1997). Standard mineral separation and $^{40}Ar/^{39}Ar$ techniques were followed (see methods) on two splits of large grains of un-weathered phlogopites. Results for stepwise degassing of the samples are shown in Fig. 2a (analytical results are given in Appendix A, errors reported are 2σ). The first split yields a well defined plateau at 1061.7 ± 9.7 Ma (66% of the gas, MSWD = 1.02) and a total fusion age of 1068.2 ± 9.4 Ma (Fig. 2a). The isochron age for split #1 is 1068.3 ± 14.0 Ma with a low MSWD of 0.46 (Fig. 2b). The second split yields an older plateau age of 1078.4 ± 11.4 Ma (82% of the gas, MSWD = 3.31) and a total fusion age of 1080 ± 10 Ma (Fig. 2a). The isochron age for split #2 is 1072.2 ± 21 Ma with an MSWD of 2.70 (Fig. 2c). Inverse isochrons for both samples show a very small spread in $^{39}Ar/^{40}Ar$ ratios and yield ages of 1066.5 ± 13 Ma and 1061.3 ± 20.2 Ma respectively. All steps are too radiogenic to utilize inverse isochron analysis in any meaningful way.

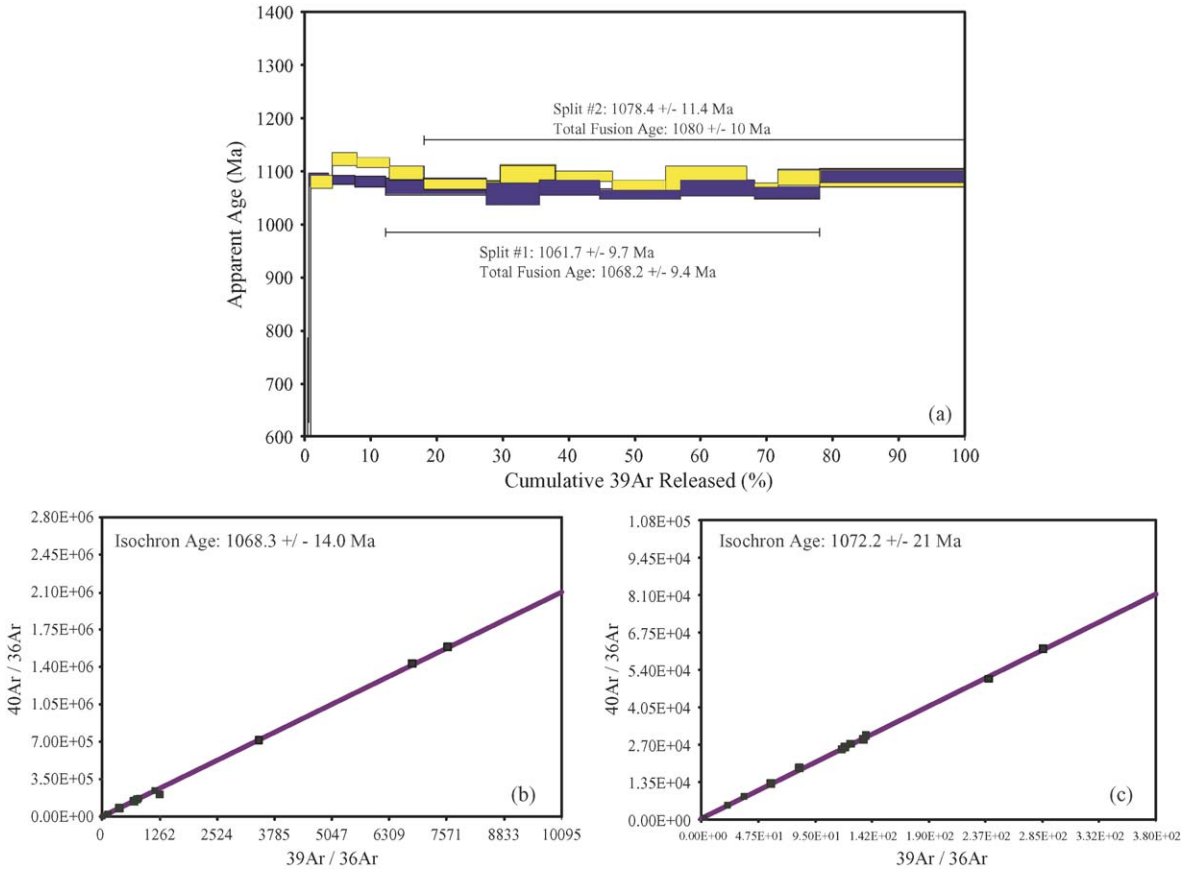


Fig. 2. (a) Stepwise degassing spectrum for Majhgawan phlogopites (2 splits). Both show well-defined plateaus that overlap at the 2σ level. The average age obtained from a weighted mean of plateau steps for splits 1 and 2 is 1073.5 ± 13.7 Ma (b) Isochron plot for the first split of phlogopite that yielded an age of 1068.3 ± 14 Ma and (c) isochron plot for the second split of phlogopite that yielded an age of 1072 ± 21 Ma.

Although the splits show a 17 Ma difference in their plateau ages, they overlap at the 2σ level. Our best estimate for the age of the phlogopite micas from Majhgawan is given by a weighted mean of the ages within the plateau segments of both samples, which is 1073.5 ± 13.7 Ma.

Kimberlites and lamproites are both thought to be emplaced into the shallow crust on time scales ranging from several hours to several days (Skinner and Marsh, 2004). Thus, the age of the phlogopites should closely date the time of upper crustal emplacement and also the age of the magnetization in the rocks.

5.2. Paleomagnetism

A total of 10 samples were collected from the brecciated kimberlite and basaltic kimberlite (eight samples exhibited stable demagnetization behavior). Typical demagnetization behavior is shown in Fig. 3 using both alternating field (Fig. 3a) and thermal (Fig. 3b) treatments. Alternating field treatments up to 60 mT removed over 90% of the natural remanant magnetization (NRM). Thermal demagnetization was applied up to temperatures of 600 °C and a loss of 90% of initial NRM strength occurred at about 550 °C with a discrete unblocking temperature range between 500 and 550 °C. The mean direction from our samples is $D = 45.1°$, $I = -25.1°$ ($k = 37$ and $\alpha_{95} = 9.3°$; Fig. 4a).

Paleomagnetic data from Majhgawan, including the data from Miller and Hargraves (1994) yield a mean direction with a declination of 37.5° and an inclination of −26.5° ($k = 66$ and $\alpha_{95} = 15.3°$; Fig. 4a) with a virtual geomagnetic pole (VGP) of 36.8°N, 212.5°E (dp = 9.0, dm = 16.6; Fig. 4c). The VGP translates to a site latitude of $14° + 8.3°/-10°$. Demagnetization and rock magnetic behavior (Figs. 3 and 5) indicate the primary carrier is magnetite. Curie temperature runs (Fig. 5a and b) show a sharp drop in susceptibility at 591.3 and 605.1 °C, which

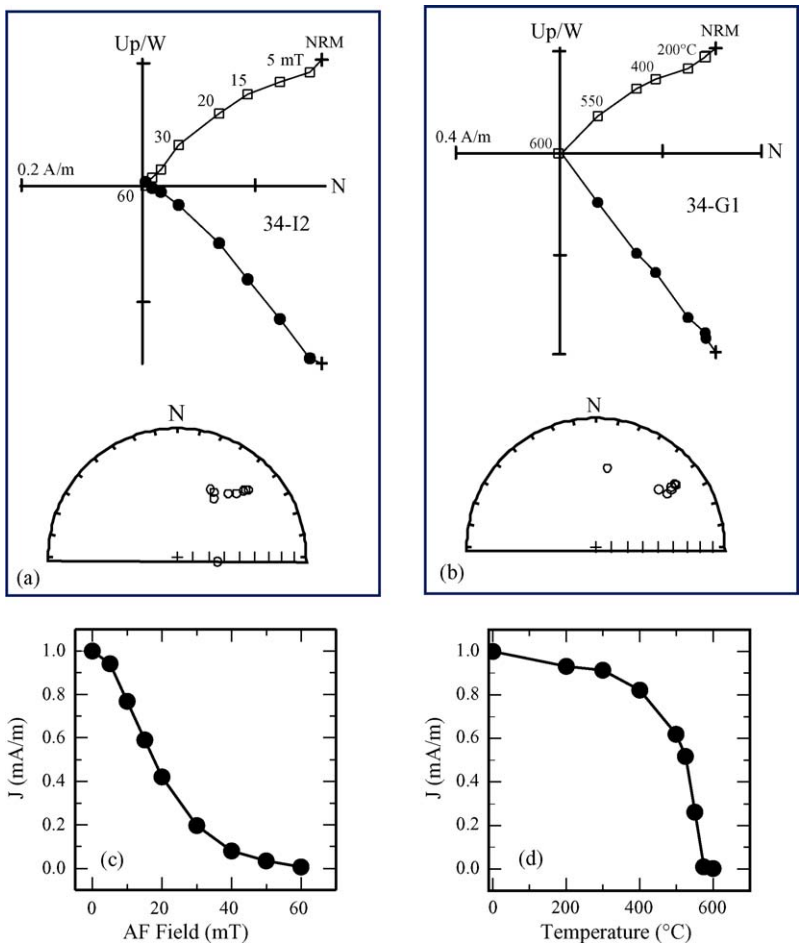


Fig. 3. (a) Alternating field demagnetization Zijderveld plot for a brecciated kimberlite sample showing univectorial behavior. (b) Thermal demagnetization Zijderveld plots for a brecciated kimberlite sample showing univectorial behavior. (c) Intensity decay plot for an alternating field demagnetized sample and (d) intensity decay plot for a thermally demagnetized sample.

are both slightly above the Curie temperature for magnetite. Isothermal remanence magnetization tests show a rapid rise in intensity and near saturation at 0.4–0.7 Telsa. The rock magnetic tests and demagnetization behavior are both characteristic of magnetite with perhaps a small contribution from hematite. Petrographic observations were described in Miller and Hargraves (1994) who noted the presence of fine magnetite grains in the matrix

Table 2
Summary of paleomagnetic results

Study	Ns (#samples)	Dec	Inc	K	α_{95}	Pole lat	Pole long	dp	dm
Majhgawan									
This paper	8	45.3°	−25.1°	37	9.3	33.5°N	203.3°E		
Miller and Hargraves (1994), site 1	6	40.2	−32.3	28	12.8	32.4°N	213.2°E		
Miller and Hargraves (1994), site 2	6	27.4	−21.6	54	9.2	45.3°N	220.1°E		
Mean of 3 sites	22	37.5°	−26.5°	66	15.3	36.8°N	212.5°E	9.0°	16.6°
Malani mean[a]	4 Studies	359.1°	62°	73.17	7.9	72.7°N	70.5°E	9.5	12.3
Bhander–Rewa mean[b]	18 Sites	34.2°	−20.3°	21.2	7	41.6°N	212.3°E	3.8°	7.2°

Ns, number of sites or #samples; Dec, declination; Inc, inclination; α_{95}, circle of 95% confidence; Pole lat, latitude of the paleomagnetic pole; Pole long, longitude of paleomagnetic pole; (dp, dm) cone of 95% confidence about the paleomagnetic pole in the co-latitude direction (dp) and at a right angle to the co-latitude direction (dm).

[a] Mean reported from Klootwijk (1975), Athavale et al. (1963), Torsvik et al. (2001) and Gregory et al. (2005).

[b] Mean reported from McElhinny et al. (1978).

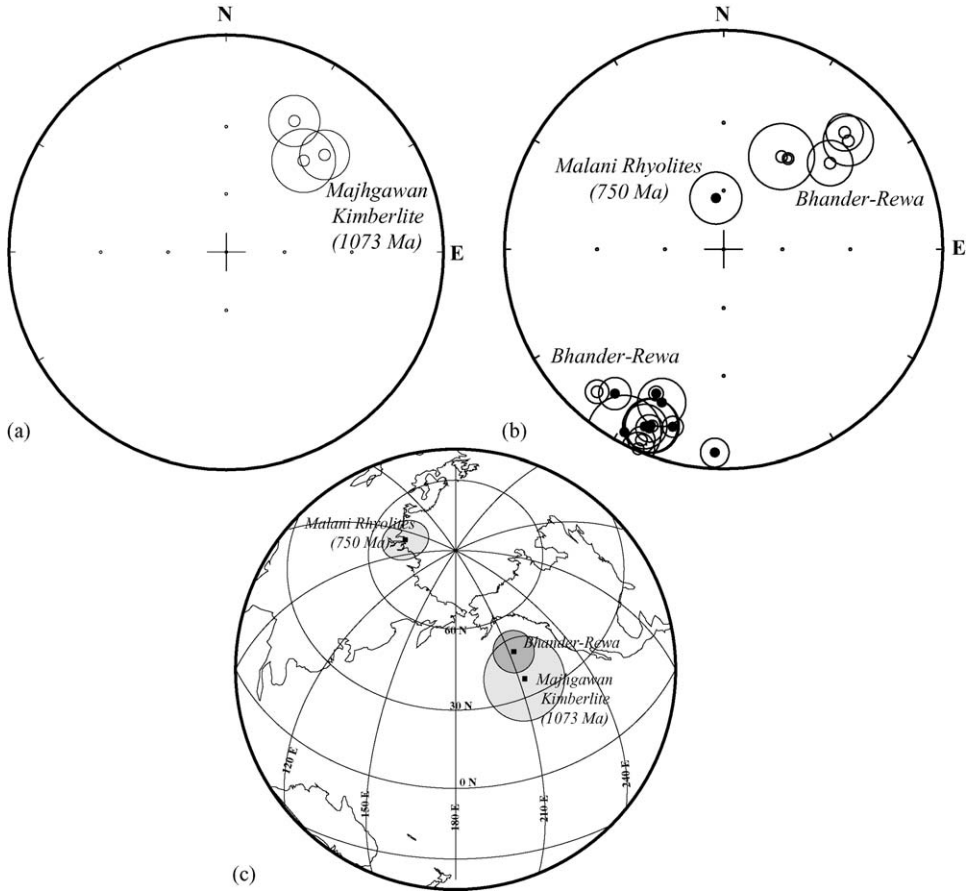


Fig. 4. (a) Site means for three sites in the Majhgawan kimberlite collected in this study and also the results of Miller and Hargraves (1994). (b) Site mean directions (tilt-corrected) from sites in the Bhander and Rewa Groups from McElhinny et al. (1978), Klootwijk (1973) and Athavale et al. (1963). We also show the combined mean pole for the Malani igneous province including results from late-stage Malani dykes reported in Gregory et al. (2005). (c) Paleomagnetic and virtual geomagnetic poles for the Bhander–Rewa Groups, the Malani Igneous province and the Majhgawan kimberlites.

and within mica flakes which in some cases had been altered to hematite. These observations are consistent with the rock magnetic data collected during our study (Table 2).

6. Discussion and age of the Upper Vindhyan

The age of the Upper Vindhyan sequence is contentious. Previous estimates on the age of Kaimur-Rewa and Bhander Groups unrealistically span over 1 billion years, from 1650 Ma to as young as Ordovician (Ray, 2006; Auden, 1933). The Bhander–Rewa Groups are at the center of this argument because they are generally considered to be Neoproterozoic in age (750–650 Ma; Ray et al., 2003; Kumar et al., 2002). The age estimates of Ray et al. (2003) and Kumar et al. (2002) are based primarily on the comparison to global curves of $^{87}Sr/^{86}Sr$ ratios and $\delta^{13}C$ dates from the Bhander and Lahkeri limestones. These ratios were used to argue for an age of Bhander sedimentation between 750 and 650 Ma. A 750 Ma age for the Bhander limestones is precluded by the assumption of a primary magnetization carried in the Bhander–Rewa rocks (McElhinny et al., 1978). India has a well-defined paleomagnetic pole at 750 Ma derived from the Malani Igneous Suite (Torsvik et al., 2001) that is distinct from the published Bhander–Rewa directions (Fig. 3). A late Neoproterozoic or Cambrian age for the Bhander–Rewa was supported by a comparison between paleomagnetic directions from Late Proterozoic–Cambrian age sedimentary rocks in the Salt Range of Pakistan (McElhinny et al., 1978; Meert, 2003; Meert and Torsvik, 2003). Although the paleomagnetic poles from the Salt Range rocks are similar to the Bhander–Rewa directions, Klootwijk et

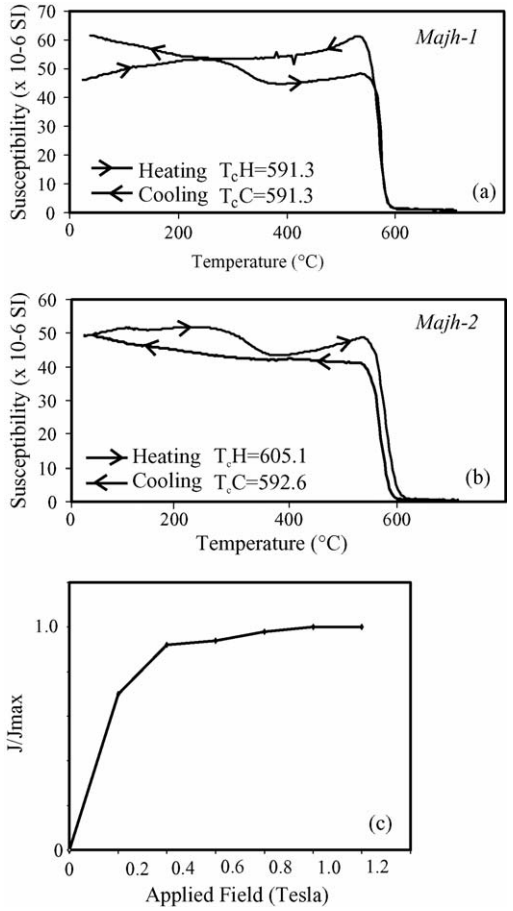


Fig. 5. (a) Temperature–susceptibility graph for a sample of the Majhgawan kimberlite showing a heating and cooling Curie temperature of 591.3 °C. (b) Temperature–susceptibility graph for a sample of the Majhgawan kimberlite showing a heating Curie temperature of 605.1 °C and a cooling Curie temperature of 592.6 °C. (c) Isothermal remanence acquisition curve for a sample of Majhgawan kimberlite showing saturation at 0.7 Tesla.

al. (1986) demonstrated significant vertical axis rotation of units in the Salt Ranges (up to 45°) during the Tertiary, negating the validity of comparisons between these units.

Additional evidence forwarded in support of a Neoproterozoic age is generally poor. The previously noted fossils from De (2003) do not provide a well-constrained date. Identification of the impressions as Ediacaran is problematic; however the more recently documented fossils (De, in press) from the same region show much better preservational characteristics and would suggest that the Lhakeri limestone is younger than ~635 Ma. Chakrabarti (1990) also documents possible trace fossils in the Bhander Group. These are noted as burrow zones with large diameter burrows (0.5–4.5 cm) and micro-burrows (less than 1.5 mm) that are lined with a thin layer of clay. Yet it is mentioned that these traces may be dubiofossils produced by either sand collapse or fluid escape. Nevertheless, Chakrabarti (1990) suggested a Riphean age (1400–800 Ma) for the Bhander–Rewa Groups. Both *Chuaria* and *Tawuia* fossils have been found in the Bhander–Rewa (Rai et al., 1997; Kumar and Srivastava, 2003), but the age range of *Chuaria* may span as far back as the Paleoproterozoic (Steiner, 1997) and thus they are not useful as index fossils for the Neoproterozoic. *Chuaria* and *Tawuia* are reported in the Suket shales of the Lower Vindhyan Semri Group (Kumar et al., 2001). Steiner (1997) gives a preferred range for the *Chuaria–Tawuia* assemblage from 1000 to 700 Ma.

Lastly, although the exact number of glaciations and the extent of those glaciations are debated, the Neoproterozoic is known for the presence of glaciogenic sequences on nearly all the continents (Evans, 2000). The Upper Vindhyan sequence shows only cryptic evidence of glaciogenic sediments despite a nearly continuous record of sedimentation (Prasad, 1984; Kumar et al., 2002). Kumar et al. (2002) noticed a large negative shift in $\delta^{13}C$ values in the Lahkeri limestone (Bhander Group, Rajasthan) and argued that it might represent a peninsular India equivalent of a cap-carbonate. Kumar et al. (2002) also mentioned the occurrence of 'tilloid' rocks beneath the Lahkeri limestone in Rajasthan as potential representatives of the Neoproterozoic 'Snowball Earth'. In contrast, the Lahkeri limestone in the Son Valley section shows no negative $\delta^{13}C$ excursion (Ray et al., 2003) suggesting either that (a) previous correlations of these units or (b) one of the carbon studies is incorrect or (c) carbon isotopic ratios show lateral variations across the basin (see also Kaufman et al., 2006). Our geochronologic data from the Majhgawan kimberlite suggests that the Bhander–Rewa Groups are both younger than ~1075 Ma.

Paleomagnetic data from our study along with previous studies of the Majhgawan kimberlite (Miller and Hargraves, 1994) and the Bhander–Rewa Groups (McElhinny et al., 1978) present an intriguing alternative interpretation for the age of the Upper Vindhyan. Prior paleomagnetic studies have been reported on the Bhander–Rewa Groups overlying the Majhgawan kimberlite (McElhinny et al., 1978; Klootwijk, 1973; Athavale et al., 1963). The paleomagnetic directions from the Bhander–Rewa yield a mean declination of 34.2° and an inclination of −20.3° ($k = 35$, $\alpha_{95} = 7°$). This compares favorably with the Majhgawan data reported in this study and that by Miller and Hargraves (1994) (Fig. 3).

There are several explanations for the similarity in directional data from the Majhgawan kimberlite and the Bhander–Rewa Groups. One option is that they are both remagnetizations of Neoproterozoic or Cambrian age. Other possibilities are (a) that the Mesoproterozoic directions in the Majhgawan kimberlite are fortuitously identical to the much younger Bhander–Rewa Group (b) the kimberlite was remagnetized during Bhander–Rewa time or (c) unrecognized tilting of the Majhgawan kimberlite has resulted in the incorrect use of the *in situ* directions. A final possibility is that the Bhander–Rewa Groups are only slightly younger than the Majhgawan kimberlite. At present, we feel comfortable rejecting the possibility of unrecognized tilting of the kimberlite. Tilting of the Kaimur sandstone away from the intrusion is minor (<10°) and therefore any post-intrusion tilting is likely to be minor as well. The other explanations require a detailed paleomagnetic study of the Bhander and Rewa Groups. We are currently in the process of completing such a study.

7. Conclusions

Our geochronologic study of Majhgawan yields a well constrained $^{40}Ar/^{39}Ar$ age of 1073.5 ± 13.7 Ma and a virtual geomagnetic pole at 36.8°N, 212.5°E (dp = 9.0, dm = 16.6) that is similar to previously published paleomagnetic poles in the overlying Bhander–Rewa Groups (47.3° N, 212.7° E; dp = 3.8°, dm = 7.2°). Our age helps constrain the age of sedimentation in the Upper Vindhyan Bhander and Rewa Groups to less than 1075 Ma.

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Appendix A

Power (Watts)	36Ar[a] (Volts)	37Ar[b] (Volts)	39Ar (Volts)	40Ar[c] (Volts)	40Ar*[d] (%)	40Ar*/39Ar	Cumulative 39Ar (%)	Calculated age (Ma)	Error in age (±2S.D.)	K/Ca
(a) Majh01 biotite; $J = .00384$[e]										
0.11 (2.1)	0.00067	0.00024	0.00126	0	0	0.00000	0.11	0	±0.00	2.26
0.21(2.3%)	0.00013	0.00061	0.00134	0.03697	49.84	27.68094	0.12	182.24	±109.22	0.949
0.32 (2.6%)	0.001	0.00792	0.00224	0	0	0.00000	0.2	0	±0.00	0.122
0.42 (3%)	0.00012	0.00147	0.00153	0.19068	84.56	124.67805	0.14	705.77	±79.65	0.448
0.74 (3.8%)	0.00026	0.00218	0.03306	7.07956	98.91	214.17018	2.94	1082.74	±12.57	6.526
0.80 (5%)	0.00004	0.00018	0.04522	9.68697	99.87	214.23648	4.02	1082.99	±8.40	110.226
0.85 (4.0%)	0.00006	0.00036	0.05248	11.18651	99.82	213.15344	4.66	1078.87	±9.84	62.358
0.95 (4.2%)	0.00022	0.00294	0.17232	36.33149	99.81	210.83660	15.32	1070.02	±14.83	25.176
1.0 (4.4%)	0.00013	0.0018	0.09053	18.76306	99.79	207.25576	8.05	1056.26	±20.53	21.625
1.06 (4.6%)	0	0.00382	0.10145	21.34383	99.99	210.38313	9.02	1068.28	±14.41	11.425
1.16 (4.9%)	0.00036	0.00368	0.13838	28.65372	99.61	207.05847	12.3	1055.5	±8.87	16.163
1.27 (5.2%)	0.00002	0.00088	0.12529	26.35047	99.97	210.30834	11.14	1068	±14.83	61.311
1.38 (5.4%)	0.00003	0.00114	0.11219	23.37715	99.95	208.38027	9.97	1060.59	±12.97	42.283
Fused	0	0.00019	0.24779	53.66914	99.99	216.59434	22.02	1091.93	±12.90	563.464
(b) Majh02 biotite; $J = 0.00384$										
0.42 (3.1%)	0.00240	0.00161	0.00655	0.05114	6.72	7.80610	0.84	53.28	±62.08	1.748

Appendix A (*Continued*)

Power (Watts)	36Ar[a] (Volts)	37Ar[b] (Volts)	39Ar (Volts)	40Ar[c] (Volts)	40Ar*[d] (%)	40Ar*/39Ar	Cumulative 39Ar (%)	Calculated age (Ma)	Error in age (±2S.D.)	K/Ca
0.74 (3.8%)	0.00120	0.00000	0.02612	5.57510	94.03	213.44767	3.36	1079.99	±11.96	0.000
0.85 (4.1%)	0.00081	0.00000	0.02899	6.52504	96.44	225.05125	3.73	1123.63	±11.78	0.000
0.95 (4.3%)	0.00046	0.00000	0.03769	8.40122	98.40	222.90544	4.85	1115.64	±9.06	0.000
1.06 (4.5%)	0.00030	0.00000	0.04095	8.91638	99.01	217.75048	5.27	1096.30	±12.24	0.000
1.16 (4.7%)	0.00077	0.00334	0.09027	19.11802	98.81	211.79462	11.61	1073.69	±8.44	11.637
1.16 (4.8%)	0.00113	0.00139	0.06515	14.16368	97.68	217.38796	8.38	1094.93	±16.83	20.122
1.16 (4.9%)	0.00054	0.00287	0.06728	14.54292	98.90	216.14813	8.65	1090.24	±9.15	10.094
1.16 (5.1%)	0.00047	0.00300	0.06300	13.28969	98.96	210.96016	8.10	1070.49	±11.77	9.016
1.38 (5.4%)	0.00033	0.00186	0.09505	20.39615	99.51	214.57669	12.22	1084.28	±24.72	21.950
1.48 (5.6%)	0.00015	0.00000	0.03697	7.76653	99.40	210.09280	4.75	1067.17	±10.00	0.139
Fused	0.00184	0.00134	0.21955	47.23523	98.85	215.15006	28.23	1086.46	±16.11	70.453

[a] Atmospheric argon.
[b] Calcium interference.
[c] Radiogenic argon.
[d] Percent of argon gas released.
[e] *J* parameter is the standard flux monitor used to normalize the amount of K converted to Ar during irradiation.

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